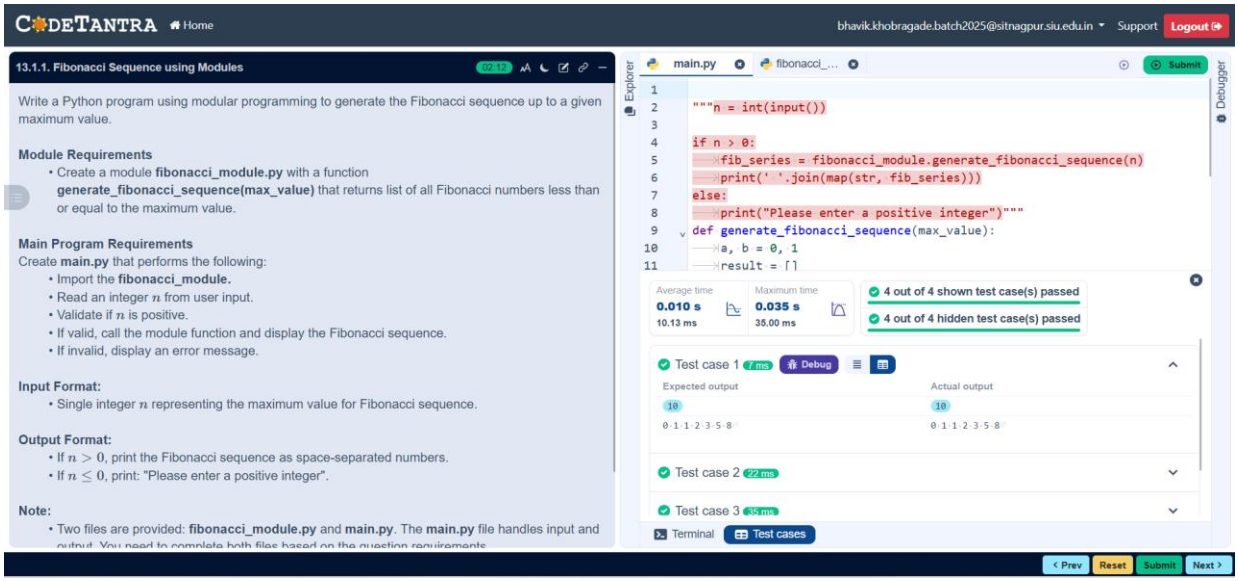
Experiment 13.1.1

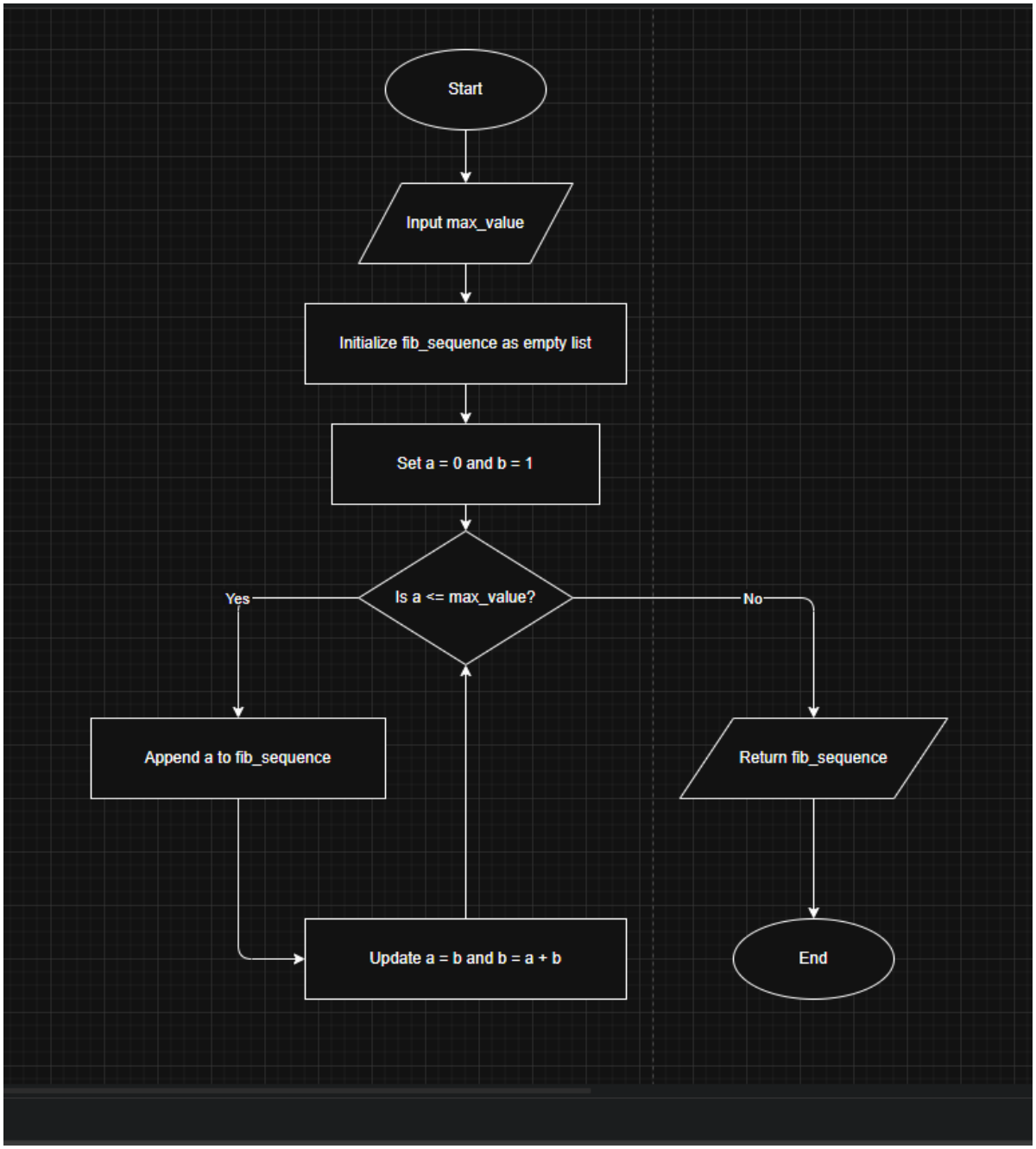
Algorithm –

1. Start
2. Input
3. Read max_value from the user.
4. Initialize
5. Create an empty list fib_sequence
6. Set a = 0 and b = 1
7. Loop Condition
8. While a ≤ max_value, repeat steps 5–6:
9. Store Value
10. Append a to fib_sequence
11. Update Values
12. Set a = b
13. Set b = a + b (using previous values)
14. Exit Loop
15. When a > max_value, stop the loop
16. Output
17. Return (or display) fib_sequence
18. End

Excecution –



Flowchart –


Start
Input max_value
Initialize fib_sequence as empty list
Set a = 0 and b = 1
Is a <= max_value?
Yes
No
Append a to fib_sequence
Return fib_sequence
Update a = b and b = a + b
End